NAME

mbmakeplatform − creates or modifies an MB-System platform file.

VERSION

Version 5.0

SYNOPSIS

mbmakeplatform
--output=*plffile*
[
--verbose
--help
--input=*plffile*
--swath=*datalist*
--swath=*swathfile*
--swath-format=*value*
--platform-type-surface-vessel
--platform-type-tow-body
--platform-type-rov
--platform-type-auv
--platform-type-aircraft
--platform-type-satellite
--platform-name=*string*
--platform-organization=*string*
--platform-documentation-url=*string*
--platform-start-time=*yyyy/mm/dd/hh/mm/ss.ssssss*
--platform-end-time=*yyyy/mm/dd/hh/mm/ss.ssssss*
--add-sensor-sonar-echosounder
--add-sensor-sonar-multiechosounder
--add-sensor-sonar-sidescan
--add-sensor-sonar-interferometry
--add-sensor-sonar-multibeam
--add-sensor-sonar-multibeam-twohead
--add-sensor-sonar-subbottom
--add-sensor-camera-mono
--add-sensor-camera-stereo
--add-sensor-camera-video
--add-sensor-lidar-scan
--add-sensor-lidar-swath
--add-sensor-position
--add-sensor-compass
--add-sensor-vru
--add-sensor-imu
--add-sensor-ins
--add-sensor-ins-with-pressure
--add-sensor-ctd
--add-sensor-pressure
--add-sensor-soundspeed
--modify-sensor=*sensorid*
--modify-sensor-bathymetry
--modify-sensor-bathymetry1
--modify-sensor-bathymetry2

--modify-sensor-bathymetry3
--modify-sensor-backscatter
--modify-sensor-backscatter1
--modify-sensor-backscatter2
--modify-sensor-backscatter3
--modify-sensor-subbottom
--modify-sensor-subbottom1
--modify-sensor-subbottom2
--modify-sensor-subbottom3
--modify-sensor-camera
--modify-sensor-camera1
--modify-sensor-camera2
--modify-sensor-camera3
--modify-sensor-position
--modify-sensor-position1
--modify-sensor-position2
--modify-sensor-position3
--modify-sensor-depth
--modify-sensor-depth1
--modify-sensor-depth2
--modify-sensor-depth3
--modify-sensor-heading
--modify-sensor-heading1
--modify-sensor-heading2
--modify-sensor-heading3
--modify-sensor-rollpitch
--modify-sensor-rollpitch1
--modify-sensor-rollpitch2
--modify-sensor-rollpitch3
--modify-sensor-heave
--modify-sensor-heave1
--modify-sensor-heave2
--modify-sensor-heave3
--sensor-model=*string*
--sensor-manufacturer=*string*
--sensor-serialnumber=*string*
--sensor-capability-position
--sensor-capability-depth
--sensor-capability-altitude
--sensor-capability-velocity
--sensor-capability-acceleration
--sensor-capability-pressure
--sensor-capability-rollpitch
--sensor-capability-heading
--sensor-capability-magneticfield
--sensor-capability-temperature
--sensor-capability-conductivity
--sensor-capability-salinity
--sensor-capability-soundspeed
--sensor-capability-gravity
--sensor-capability-topography-echosounder
--sensor-capability-topography-interferometry
--sensor-capability-topography-sass

--sensor-capability-topography-multibeam
--sensor-capability-topography-photogrammetry
--sensor-capability-topography-structurefrommotion
--sensor-capability-topography-lidar
--sensor-capability-topography-structuredlight
--sensor-capability-topography-laserscanner
--sensor-capability-backscatter-echosounder
--sensor-capability-backscatter-sidescan
--sensor-capability-backscatter-interferometry
--sensor-capability-backscatter-sass
--sensor-capability-backscatter-multibeam
--sensor-capability-backscatter-lidar
--sensor-capability-backscatter-structuredlight
--sensor-capability-backscatter-laserscanner
--sensor-capability-photography
--sensor-capability-stereophotography
--sensor-capability-video
--sensor-capability-stereovideo
--sensor-capability1=*value*
--sensor-capability2=*value*
--sensor-offsets=*x/y/z/azimuth/roll/pitch*
--sensor-offset-positions=*x/y/z*
--sensor-offset-angles=*azimuth/roll/pitch*
--sensor-time-latency=*value*
--sensor-time-latency-model=*file*
--sensor-flipsign-heading
--sensor-flipsign-roll
--sensor-flipsign-pitch
--sensor-source-bathymetry
--sensor-source-bathymetry1
--sensor-source-bathymetry2
--sensor-source-bathymetry3
--sensor-source-backscatter
--sensor-source-backscatter1
--sensor-source-backscatter2
--sensor-source-backscatter3
--sensor-source-subbottom
--sensor-source-subbottom1
--sensor-source-subbottom2
--sensor-source-subbottom3
--sensor-source-camera
--sensor-source-camera1
--sensor-source-camera2
--sensor-source-camera3
--sensor-source-position
--sensor-source-position1
--sensor-source-position2
--sensor-source-position3
--sensor-source-depth
--sensor-source-depth1
--sensor-source-depth2
--sensor-source-depth3
--sensor-source-heading

--sensor-source-heading1
--sensor-source-heading2
--sensor-source-heading3
--sensor-source-rollpitch
--sensor-source-rollpitch1
--sensor-source-rollpitch2
--sensor-source-rollpitch3
--sensor-source-heave
--sensor-source-heave1
--sensor-source-heave2
--sensor-source-heave3
--modify-offsets=*ioff/x/y/z/azimuth/roll/pitch*
--modify-offset-positions=*ioff/x/y/z*
--modify-offset-angles=*ioff/azimuth/roll/pitch*
--modify-time-latency=*value*
--modify-time-latency-model=*file*
--end-sensor
--set-source-bathymetry=*sensorid*
--set-source-bathymetry1=*sensorid*
--set-source-bathymetry2=*sensorid*
--set-source-bathymetry3=*sensorid*
--set-source-backscatter=*sensorid*
--set-source-backscatter1=*sensorid*
--set-source-backscatter2=*sensorid*
--set-source-backscatter3=*sensorid*
--set-source-subbottom=*sensorid*
--set-source-subbottom1=*sensorid*
--set-source-subbottom2=*sensorid*
--set-source-subbottom3=*sensorid*
--set-source-camera=*sensorid*
--set-source-camera1=*sensorid*
--set-source-camera2=*sensorid*
--set-source-camera3=*sensorid*
--set-source-position=*sensorid*
--set-source-position1=*sensorid*
--set-source-position2=*sensorid*
--set-source-position3=*sensorid*
--set-source-depth=*sensorid*
--set-source-depth1=*sensorid*
--set-source-depth2=*sensorid*
--set-source-depth3=*sensorid*
--set-source-heading=*sensorid*
--set-source-heading1=*sensorid*
--set-source-heading2=*sensorid*
--set-source-heading3=*sensorid*
--set-source-rollpitch=*sensorid*
--set-source-rollpitch1=*sensorid*
--set-source-rollpitch2=*sensorid*
--set-source-rollpitch3=*sensorid*
--set-source-heave=*sensorid*
--set-source-heave1=*sensorid*
--set-source-heave2=*sensorid*
--set-source-heave3=*sensorid]*

DESCRIPTION

Mbmakeplatform is used to create an **MB-System** platform file, which provides a complete description of the geometry of sensors on a survey platform, including the relative positional and angular offsets. Platform files are used by **mbpreprocess** to set up an **MB-System** processing structure for swath data. In many cases the preprocessing consists only of creating the ancillary files parallel to the raw files, but in others the data must be translated to different formats. The preprocessing step can also be used to merge navigation, attitude, sound speed, or other ancillary data with the survey data.

Platform files are needed for preprocessing only when survey data have been collected without integrating the asynchronous navigation and attitude data and when the positional and angular offsets between the survey sensor (e.g. multibeam sonar, lidar, stereo camera rig) have not been specified in the various data streams. Most often this situation is associated with custom survey systems on submerged platforms like autonomous underwater vehicles (AUVs) or remotely operated vehicles (ROVs). The few data formats supported with preprocessing functionality utilizing platform files include the Reson 7k format 88, the Teledyne 7k3 format 89, and the 3D at Depth lidar formats 232 and 233.

Generally platform files are created by a single execution of **Mbmakeplatform** with a long series of command line arguments that add sensors, set sensor capabilities, and set sensor positional and angular offsets. For some data formats **Mbmakeplatform** can initialize a platform structure by extracting the sensor offset values from swath data files. In most cases the source of position data is set as the platform origin with zero positional and angular offset values.

No platform file is needed for preprocessing when the logged datastream includes a full platform description. Specifically, all datasets collected using Kongsberg multibeam sonars do not require use of a platform file (the only exception to this would be if the positional offsets between sensors were incorrectly specified in the multibeam configuration).

Two examples are presented below, both of which are custom submerged survey platforms combining a number of both mapping and ancillary sensors, including an inertial navigation system producing navigation and attitude.

MB-SYSTEM AUTHORSHIP

David W. Caress
Monterey Bay Aquarium Research Institute
Dale N. Chayes
Center for Coastal and Ocean Mapping
University of New Hampshire
Christian do Santos Ferreira
MARUM - Center for Marine Environmental Sciences
University of Bremen

OPTIONS

--output=*plffile*
Specifies the path of the output platform file.

[

--verbose
By default **mbmakeplatform** outputs minimal information to the shell. This option causes the program to indicate it's progress as it runs.

--help Lists the available command options.

--input=*plffile*
Specify an existing platform file to read and then modify with subsequent commands.

--swath=[]*datalist* or *swathfile*]
Specify a swath data input (a datalist file or an individual data file) from which **mbmakeplatform** will attempt to extract a starting platform structure, including the list of sensors and the sensor positional and angular offsets. If the input path name follows the **MB-System** convention (e.g. *.mb59 for format 59) or is otherwise recognized as associated with a known format, then the format need not be specified separately with **--swath-format**. Only the first available definition of the platform structure will be parsed, typically from the beginning of the first file referenced. This option only works for recent Kongsberg or Teledyne data formats (e.g. Kongsberg formats 58 & 59 and Teledyne formats 88 & 89).

--swath-format=*value*
Specify the format of the swath file or datalist from which a starting platform structure should be extracted.

--platform-type-surface-vessel
Set the platform type to be a surface vessel.

--platform-type-tow-body
Set the platform type to be a towed body.

--platform-type-rov
Set the platform type to be an ROV (remotely operated vehicle).

--platform-type-auv
Set the platform type to be an AUV (autonomous underwater vehicle).

--platform-type-aircraft
Set the platform type to be an aircraft.

--platform-type-satellite
Set the platform type to be a satellite.

--platform-name=*string*
Set the platform name.

--platform-organization=*string*
Set the name of the platform operating organization.

--platform-documentation-url=*string*
Set the name of the platform documentation website url.

--platform-start-time=*yyyy/mm/dd/hh/mm/ss.ssssss*
Set the starting time for the time interval that this platform description is valid.

--platform-end-time=*yyyy/mm/dd/hh/mm/ss.ssssss*
Set the ending time for the time interval that this platform description is valid.

--add-sensor-sonar-echosounder
Initiate adding a new sensor of type echosounder (sonar). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-sonar-multiechosounder
Initiate adding a new sensor of type multiechosounder (sonar). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-sonar-sidescan
Initiate adding a new sensor of type sidescan (sonar). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-sonar-interferometry
Initiate adding a new sensor of type interferometry sidescan (sonar). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-sonar-multibeam
Initiate adding a new sensor of type multibeam (sonar). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-sonar-multibeam-twohead
Initiate adding a new sensor of type two-head multibeam (sonar). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-sonar-subbottom
Initiate adding a new sensor of type subbottom (sonar). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-camera-mono
Initiate adding a new sensor of type single still camera. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-camera-stereo
Initiate adding a new sensor of type stereo still camera rig. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-camera-video
Initiate adding a new sensor of type video camera. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-lidar-scan
Initiate adding a new sensor of type 2D scanning lidar. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-lidar-swath
Initiate adding a new sensor of type swath lidar. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-position
Initiate adding a new sensor of type position. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-compass
Initiate adding a new sensor of type compass. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-vru
Initiate adding a new sensor of type vertical reference unit (VRU). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-imu
Initiate adding a new sensor of type inertial motion unit (IMU). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-ins
Initiate adding a new sensor of type inertial navigation system (INS). All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-ins-with-pressure
Initiate adding a new sensor of type inertial navigation system (INS) with pressure. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-ctd
Initiate adding a new sensor of type CTD. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-pressure
Initiate adding a new sensor of type pressure. All commands will pertain to this sensor until the command **end-sensor** is given.

--add-sensor-soundspeed
Initiate adding a new sensor of type sound speed. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor=*sensorid*
Initiate modifying the existing sensor *sensorid* (where sensors ids count from 0). All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-bathymetry
Initiate modifying the existing sensor that is identified as the source for bathymetry. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-bathymetry1
Initiate modifying the existing sensor that is identified as the source for bathymetry1. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-bathymetry2
Initiate modifying the existing sensor that is identified as the source for bathymetry2. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-bathymetry3
Initiate modifying the existing sensor that is identified as the source for bathymetry3. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-backscatter
Initiate modifying the existing sensor that is identified as the source for backscatter. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-backscatter1
Initiate modifying the existing sensor that is identified as the source for backscatter1. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-backscatter2
Initiate modifying the existing sensor that is identified as the source for backscatter2. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-backscatter3
Initiate modifying the existing sensor that is identified as the source for backscatter3. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-subbottom
Initiate modifying the existing sensor that is identified as the source for subbottom. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-subbottom1
Initiate modifying the existing sensor that is identified as the source for subbottom1. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-subbottom2
Initiate modifying the existing sensor that is identified as the source for subbottom2. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-subbottom3
Initiate modifying the existing sensor that is identified as the source for subbottom3. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-camera
Initiate modifying the existing sensor that is identified as the source for camera. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-camera1
Initiate modifying the existing sensor that is identified as the source for camera1. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-camera2
Initiate modifying the existing sensor that is identified as the source for camera2. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-camera3
Initiate modifying the existing sensor that is identified as the source for camera3. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-position
Initiate modifying the existing sensor that is identified as the source for position. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-position1
Initiate modifying the existing sensor that is identified as the source for position1. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-position2
Initiate modifying the existing sensor that is identified as the source for position2. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-position3
Initiate modifying the existing sensor that is identified as the source for position3. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-depth
Initiate modifying the existing sensor that is identified as the source for depth. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-depth1
Initiate modifying the existing sensor that is identified as the source for depth1. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-depth2
Initiate modifying the existing sensor that is identified as the source for depth2. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-depth3
Initiate modifying the existing sensor that is identified as the source for depth3. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-heading
Initiate modifying the existing sensor that is identified as the source for heading. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-heading1
Initiate modifying the existing sensor that is identified as the source for heading1. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-heading2
Initiate modifying the existing sensor that is identified as the source for heading2. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-heading3
Initiate modifying the existing sensor that is identified as the source for heading3. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-rollpitch
Initiate modifying the existing sensor that is identified as the source for rollpitch. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-rollpitch1
Initiate modifying the existing sensor that is identified as the source for rollpitch1. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-rollpitch2
Initiate modifying the existing sensor that is identified as the source for rollpitch2. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-rollpitch3
Initiate modifying the existing sensor that is identified as the source for rollpitch3. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-heave
Initiate modifying the existing sensor that is identified as the source for heave. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-heave1
Initiate modifying the existing sensor that is identified as the source for heave1. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-heave2
Initiate modifying the existing sensor that is identified as the source for heave2. All commands will pertain to this sensor until the command **end-sensor** is given.

--modify-sensor-heave3
Initiate modifying the existing sensor that is identified as the source for heave3. All commands will pertain to this sensor until the command **end-sensor** is given.

--sensor-model=*string*
Set sensor model string.

--sensor-manufacturer=*string*
Set sensor manufacturer string.

--sensor-serialnumber=*string*
Set sensor serial number string.

--sensor-capability-position
Add position to list of sensor’s measurement capabilities.

--sensor-capability-depth
Add depth (typically from pressure) to list of sensor’s measurement capabilities.

--sensor-capability-altitude
Add altitude to list of sensor’s measurement capabilities.

--sensor-capability-velocity
Add velocity of motion to list of sensor’s measurement capabilities.

--sensor-capability-acceleration
Add acceleration of motion to list of sensor's measurement capabilities.

--sensor-capability-pressure
Add pressure to list of sensor's measurement capabilities.

--sensor-capability-rollpitch
Add roll and pitch to list of sensor's measurement capabilities.

--sensor-capability-heading
Add heading to list of sensor's measurement capabilities.

--sensor-capability-magneticfield
Add magnetic field to list of sensor's measurement capabilities.

--sensor-capability-temperature
Add temperature to list of sensor's measurement capabilities.

--sensor-capability-conductivity
Add conductivity to list of sensor's measurement capabilities.

--sensor-capability-salinity
Add salinity to list of sensor's measurement capabilities.

--sensor-capability-soundspeed
Add sound speed to list of sensor's measurement capabilities.

--sensor-capability-gravity
Add gravity to list of sensor's measurement capabilities.

--sensor-capability-topography-echosounder
Add topography from echosounder to list of sensor's measurement capabilities.

--sensor-capability-topography-interferometry
Add topography from interferometric sidescan sonar to list of sensor's measurement capabilities.

--sensor-capability-topography-sass
Add topography from synthetic aperture sidescan (SASS) sonar to list of sensor's measurement capabilities.

--sensor-capability-topography-multibeam
Add topography from multibeam sonar to list of sensor's measurement capabilities.

--sensor-capability-topography-photogrammetry
Add topography from photogrammetry to list of sensor's measurement capabilities.

--sensor-capability-topography-structurefrommotion
Add topography from structure from motion to list of sensor's measurement capabilities.

--sensor-capability-topography-lidar
Add topography from lidar to list of sensor's measurement capabilities.

--sensor-capability-topography-structuredlight
Add topography from structured light to list of sensor's measurement capabilities.

--sensor-capability-topography-laserscanner
Add topography from laser scanner to list of sensor's measurement capabilities.

--sensor-capability-backscatter-echosounder
Add backscatter from echosounder to list of sensor's measurement capabilities.

--sensor-capability-backscatter-sidescan
Add backscatter from sidescan sonar to list of sensor's measurement capabilities.

--sensor-capability-backscatter-interferometry
Add backscatter from interferometric sidescan sonar to list of sensor's measurement capabilities.

--sensor-capability-backscatter-sass
Add backscatter from synthetic aperture sidescan (SASS) sonar to list of sensor's measurement capabilities.

--sensor-capability-backscatter-multibeam
Add backscatter from multibeam sonar to list of sensor's measurement capabilities.

--sensor-capability-backscatter-lidar
Add backscatter from lidar to list of sensor's measurement capabilities.

--sensor-capability-backscatter-structuredlight
Add backscatter from structured light to list of sensor's measurement capabilities.

--sensor-capability-backscatter-laserscanner
Add backscatter from laser scanner to list of sensor's measurement capabilities.

--sensor-capability-photography
Add single camera still photography to list of sensor's measurement capabilities.

--sensor-capability-stereophotography
Add stereo camera still photography to list of sensor's measurement capabilities.

--sensor-capability-video
Add video to list of sensor's measurement capabilities.

--sensor-capability-stereovideo
Add stereo video to list of sensor's measurement capabilities.

--sensor-capability1=*value*
Set the first of two sensor capability 4 byte bitmasks directly. The usage of the bits in this mask are:

MB_SENSOR_CAPABILITY1_NONE 0x00000000 All bits = 0
MB_SENSOR_CAPABILITY1_POSITION 0x00000001 Bit 0 = 1
MB_SENSOR_CAPABILITY1_DEPTH 0x00000002 Bit 1 = 2
MB_SENSOR_CAPABILITY1_ALTITUDE 0x00000004 Bit 2 = 4
MB_SENSOR_CAPABILITY1_VELOCITY 0x00000008 Bit 3 = 8
MB_SENSOR_CAPABILITY1_ACCELERATION 0x00000010 Bit 4 = 16
MB_SENSOR_CAPABILITY1_PRESSURE 0x00000020 Bit 5 = 32
MB_SENSOR_CAPABILITY1_ROLLPITCH 0x00000040 Bit 6 = 64
MB_SENSOR_CAPABILITY1_HEADING 0x00000080 Bit 7 = 128
MB_SENSOR_CAPABILITY1_HEAVE 0x00000100 Bit 8 = 256
MB_SENSOR_CAPABILITY1_UNUSED09 0x00000200 Bit 9 = 512
MB_SENSOR_CAPABILITY1_UNUSED10 0x00000400 Bit 10 = 1024
MB_SENSOR_CAPABILITY1_UNUSED11 0x00000800 Bit 11 = 2048

MB_SENSOR_CAPABILITY1_UNUSED12
0x00001000 Bit 12 = 4096
MB_SENSOR_CAPABILITY1_TEMPERATURE
0x00002000 Bit 13 = 8192
MB_SENSOR_CAPABILITY1_CONDUCTIVITY
0x00004000 Bit 14 = 16384
MB_SENSOR_CAPABILITY1_SALINITY
0x00008000 Bit 15 = 32768
MB_SENSOR_CAPABILITY1_SOUNDSPEED
0x00010000 Bit 16 = 65536
MB_SENSOR_CAPABILITY1_UNUSED17
0x00020000 Bit 17= 131072
MB_SENSOR_CAPABILITY1_UNUSED18
0x00040000 Bit 18 = 262144
MB_SENSOR_CAPABILITY1_UNUSED19
0x00080000 Bit 19 = 524288
MB_SENSOR_CAPABILITY1_GRAVITY
0x00100000 Bit 20 = 1048576
MB_SENSOR_CAPABILITY1_UNUSED21
0x00200000 Bit 21 = 2097152
MB_SENSOR_CAPABILITY1_UNUSED22
0x00400000 Bit 22 = 4194304
MB_SENSOR_CAPABILITY1_UNUSED23
0x00800000 Bit 23 = 8388608
MB_SENSOR_CAPABILITY1_MAGNETICFIELD
0x01000000 Bit 24 = 16777216
MB_SENSOR_CAPABILITY1_UNUSED25
0x02000000 Bit 25 = 33554432
MB_SENSOR_CAPABILITY1_UNUSED26
0x04000000 Bit 26 = 67108864
MB_SENSOR_CAPABILITY1_UNUSED27
0x08000000 Bit 27 = 134217728
MB_SENSOR_CAPABILITY1_UNUSED28
0x10000000 Bit 28 = 268435456
MB_SENSOR_CAPABILITY1_UNUSED29
0x20000000 Bit 29 = 536870912
MB_SENSOR_CAPABILITY1_UNUSED30
0x40000000 Bit 30 = 1073741824
MB_SENSOR_CAPABILITY1_UNUSED31
0x80000000 Bit 31 = 2147483648

--sensor-capability2=*value*

Set the second of two sensor capability 4 byte bitmasks directly. The usage of the bits in this mask are:

MB_SENSOR_CAPABILITY2_NONE
0x00000000 All bits = 0
MB_SENSOR_CAPABILITY2_TOPOGRAPHY_ECHOSOUNDER
0x00000001 Bit 0 = 1
MB_SENSOR_CAPABILITY2_TOPOGRAPHY_INTERFEROMETRY
0x00000002 Bit 1 = 2
MB_SENSOR_CAPABILITY2_TOPOGRAPHY_SASS
0x00000004 Bit 2 = 4
MB_SENSOR_CAPABILITY2_TOPOGRAPHY_MULTIBEAM

0x00000008 Bit 3 = 8
MB_SENSOR_CAPABILITY2_TOPOGRAPHY_PHOTOGRAMMETRY
0x00000010 Bit 4 = 16
MB_SENSOR_CAPABILITY2_TOPOGRAPHY_STRUCTUREFROMMOTION
0x00000020 Bit 5 = 32
MB_SENSOR_CAPABILITY2_TOPOGRAPHY_LIDAR
0x00000040 Bit 6 = 64
MB_SENSOR_CAPABILITY2_TOPOGRAPHY_STRUCTUREDLIGHT
0x00000080 Bit 7 = 128
MB_SENSOR_CAPABILITY2_TOPOGRAPHY_LASERSCANNER
0x00000100 Bit 8 = 256
MB_SENSOR_CAPABILITY2_UNUSED09
0x00000200 Bit 9 = 512
MB_SENSOR_CAPABILITY2_UNUSED10
0x00000400 Bit 10 = 1024
MB_SENSOR_CAPABILITY2_UNUSED11
0x00000800 Bit 11 = 2048
MB_SENSOR_CAPABILITY2_BACKSCATTER_ECHOSOUNDER
0x00001000 Bit 12 = 4096
MB_SENSOR_CAPABILITY2_BACKSCATTER_SIDESCAN
0x00002000 Bit 13 = 8192
MB_SENSOR_CAPABILITY2_BACKSCATTER_INTERFEROMETRY
0x00004000 Bit 14 = 16384
MB_SENSOR_CAPABILITY2_BACKSCATTER_SASS
0x00008000 Bit 15 = 32768
MB_SENSOR_CAPABILITY2_BACKSCATTER_MULTIBEAM
0x00010000 Bit 16 = 65536
MB_SENSOR_CAPABILITY2_BACKSCATTER_LIDAR
0x00020000 Bit 17= 131072
MB_SENSOR_CAPABILITY2_BACKSCATTER_STRUCTUREDLIGHT
0x00040000 Bit 18 = 262144
MB_SENSOR_CAPABILITY2_BACKSCATTER_LASERSCANNER
0x00080000 Bit 19 = 524288
MB_SENSOR_CAPABILITY2_UNUSED20
0x00100000 Bit 20 = 1048576
MB_SENSOR_CAPABILITY2_SUBBOTTOM_ECHOSOUNDER
0x00200000 Bit 21 = 2097152
MB_SENSOR_CAPABILITY2_SUBBOTTOM_CHIRP
0x00400000 Bit 22 = 4194304
MB_SENSOR_CAPABILITY2_UNUSED23
0x00800000 Bit 23 = 8388608
MB_SENSOR_CAPABILITY2_PHOTOGRAPHY
0x01000000 Bit 24 = 16777216
MB_SENSOR_CAPABILITY2_STEREOPHOTOGRAPHY
0x02000000 Bit 25 = 33554432
MB_SENSOR_CAPABILITY2_VIDEO
0x04000000 Bit 26 = 67108864
MB_SENSOR_CAPABILITY2_STEREOVIDEO
0x08000000 Bit 27 = 134217728
MB_SENSOR_CAPABILITY2_UNUSED28
0x10000000 Bit 28 = 268435456
MB_SENSOR_CAPABILITY2_UNUSED29
0x20000000 Bit 29 = 536870912

MB_SENSOR_CAPABILITY2_UNUSED30
0x40000000 Bit 30 = 1073741824
MB_SENSOR_CAPABILITY2_UNUSED31
0x80000000 Bit 31 = 2147483648

--sensor-offsets=*x/y/z/azimuth/roll/pitch*
Used to set the positional (x, y, and z) and angular (azimuth, roll, and pitch) offsets for the sensor currently being added. To modify offsets of an existing sensor use **--modify-offsets**. Here x is positive to starboard, y is positive forward, and z is positive up. Roll is positive starboard up, pitch is positive forward up, and azimuth is positive clockwise from forward. Generally one sensor has zero offsets and thus serves as the platform origin. For sensors with two elements (e.g. bistatic multibeam sonars with separate transmit and receive arrays, or stereo camera rigs with two cameras), two **--sensor-offsets** calls are made. In the case of a multibeam the first set of offsets is for the transmit array, and in the case of a stereo camera rig the first offsets are for the left camera.

--sensor-offset-positions=*x/y/z*
Used to set the positional (x, y, and z) offsets for the sensor currently being added. To modify positional offsets of an existing sensor use **--modify-offset-positions**. Here x is positive to starboard, y is positive forward, and z is positive up. Generally one sensor has zero offsets and thus serves as the platform origin. This command is used for sensors that do not make measurements dependent on orientation, and so do not have angular offsets (e.g. sound speed, pressure, or CTD sensors).

--sensor-offset-angles=*azimuth/roll/pitch*
Used to set the angular (azimuth, roll, and pitch) offsets for the sensor currently being added. To modify angular offsets of an existing sensor use **--modify-offset-angles**. Here roll is positive starboard up, pitch is positive forward up, and azimuth is positive clockwise from forward. Generally one sensor has zero offsets and thus serves as the platform origin. This command is used for sensors that do not make measurements dependent on position, and so do not have positional offsets (e.g. vertical reference units (VRUs) that measure orientation only).

--sensor-time-latency=*value*
Sets a static time latency in seconds for the sensor currently being added.

--sensor-time-latency-model=*file*
Specifies a file containing a time latency model time series for the sensor currently being added. The file is in the form of two white-space delimited columns with the first column being times in epoch seconds (seconds since the start of 1970) and the second being the time latency in seconds.

--sensor-source-bathymetry
Sets the sensor currently being either added or modified to be the source for bathymetry.

--sensor-source-bathymetry1
Sets the sensor currently being either added or modified to be the source for bathymetry1.

--sensor-source-bathymetry2
Sets the sensor currently being either added or modified to be the source for bathymetry2.

--sensor-source-bathymetry3
Sets the sensor currently being either added or modified to be the source for bathymetry3.

--sensor-source-backscatter
Sets the sensor currently being either added or modified to be the source for backscatter.

--sensor-source-backscatter1
Sets the sensor currently being either added or modified to be the source for backscatter1.

--sensor-source-backscatter2
Sets the sensor currently being either added or modified to be the source for backscatter2.

--sensor-source-backscatter3
Sets the sensor currently being either added or modified to be the source for backscatter3.

--sensor-source-subbottom
Sets the sensor currently being either added or modified to be the source for subbottom.

--sensor-source-subbottom1
Sets the sensor currently being either added or modified to be the source for subbottom1.

--sensor-source-subbottom2
Sets the sensor currently being either added or modified to be the source for subbottom2.

--sensor-source-subbottom3
Sets the sensor currently being either added or modified to be the source for subbottom3.

--sensor-source-camera
Sets the sensor currently being either added or modified to be the source for camera.

--sensor-source-camera1
Sets the sensor currently being either added or modified to be the source for camera1.

--sensor-source-camera2
Sets the sensor currently being either added or modified to be the source for camera2.

--sensor-source-camera3
Sets the sensor currently being either added or modified to be the source for camera3.

--sensor-source-position
Sets the sensor currently being either added or modified to be the source for position.

--sensor-source-position1
Sets the sensor currently being either added or modified to be the source for position1.

--sensor-source-position2
Sets the sensor currently being either added or modified to be the source for position2.

--sensor-source-position3
Sets the sensor currently being either added or modified to be the source for position3.

--sensor-source-depth
Sets the sensor currently being either added or modified to be the source for depth.

--sensor-source-depth1
Sets the sensor currently being either added or modified to be the source for depth1.

--sensor-source-depth2
Sets the sensor currently being either added or modified to be the source for depth2.

--sensor-source-depth3
Sets the sensor currently being either added or modified to be the source for depth3.

--sensor-source-heading
Sets the sensor currently being either added or modified to be the source for heading.

--sensor-source-heading1
Sets the sensor currently being either added or modified to be the source for heading1.

--sensor-source-heading2
Sets the sensor currently being either added or modified to be the source for heading2.

--sensor-source-heading3
Sets the sensor currently being either added or modified to be the source for heading3.

--sensor-source-rollpitch
Sets the sensor currently being either added or modified to be the source for rollpitch.

--sensor-source-rollpitch1
Sets the sensor currently being either added or modified to be the source for rollpitch1.

--sensor-source-rollpitch2
Sets the sensor currently being either added or modified to be the source for rollpitch2.

--sensor-source-rollpitch3
Sets the sensor currently being either added or modified to be the source for rollpitch3.

--sensor-source-heave
Sets the sensor currently being either added or modified to be the source for heave.

--sensor-source-heave1
Sets the sensor currently being either added or modified to be the source for heave1.

--sensor-source-heave2
Sets the sensor currently being either added or modified to be the source for heave2.

--sensor-source-heave3
Sets the sensor currently being either added or modified to be the source for heave3.

--end-sensor
Ends the commands for adding or modifying a sensor.

--modify-offsets=*ioff/x/y/z/azimuth/roll/pitch*
Used to set the positional (x, y, and z) and angular (azimuth, roll, and pitch) offsets for the sensor currently being modified. To add offsets to a new sensor use **--sensor-offsets**. Here x is positive to starboard, y is positive forward, and z is positive up. Roll is positive starboard up, pitch is positive forward up, and azimuth is positive clockwise from forward. Generally one sensor has zero offsets and thus serves as the platform origin. For sensors with two elements (e.g. bistatic multibeam sonars with separate transmit and receive arrays, or stereo camera rigs with two cameras), two **--sensor-offsets** calls are made. In the case of a multibeam the first set of offsets is for the transmit array, and in the case of a stereo camera rig the first offsets are for the left camera.

--modify-offset-positions=*ioff/x/y/z*
Used to set the positional (x, y, and z) offsets for the sensor currently being modified. To add positional offsets to a new sensor use **--sensor-offset-positions**. Here x is positive to starboard, y is positive forward, and z is positive up. Generally one sensor has zero offsets and thus serves as the platform origin. This command is used for sensors that do not make measurements dependent on orientation, and so do not have angular offsets (e.g. sound speed, pressure, or CTD sensors).

--modify-offset-angles=*ioff/azimuth/roll/pitch*
Used to set the angular (azimuth, roll, and pitch) offsets for the sensor currently being modified. To add angular offsets to a new sensor use **--sensor-offset-angles**. Here roll is positive starboard up, pitch is positive forward up, and azimuth is positive clockwise from forward. Generally one sensor has zero offsets and thus serves as the platform origin. This command is used for sensors that do not make measurements dependent on position, and so do not have positional offsets (e.g. vertical reference units (VRUs) that measure orientation only).

--modify-time-latency=*value*
Sets a static time latency in seconds for the sensor currently being modified.

--modify-time-latency-model=*file*
Specifies a file containing a time latency model time series for the sensor currently being modified. The file is in the form of two white-space delimited columns with the first column being times in epoch seconds (seconds since the start of 1970) and the second being the time latency in seconds.

--set-source-bathymetry=*sensorid*
Set the specified sensor to be a source for bathymetry.

--set-source-bathymetry1=*sensorid*
Set the specified sensor to be a source for bathymetry1.

--set-source-bathymetry2=*sensorid*
Set the specified sensor to be a source for bathymetry2.

--set-source-bathymetry3=*sensorid*
Set the specified sensor to be a source for bathymetry3.

--set-source-backscatter=*sensorid*
Set the specified sensor to be a source for backscatter.

--set-source-backscatter1=*sensorid*
Set the specified sensor to be a source for backscatter1.

--set-source-backscatter2=*sensorid*
Set the specified sensor to be a source for backscatter2.

--set-source-backscatter3=*sensorid*
Set the specified sensor to be a source for backscatter2.

--set-source-subbottom=*sensorid*
Set the specified sensor to be a source for subbottom.

--set-source-subbottom1=*sensorid*
Set the specified sensor to be a source for subbottom1.

--set-source-subbottom2=*sensorid*
Set the specified sensor to be a source for subbottom2.

--set-source-subbottom3=*sensorid*
Set the specified sensor to be a source for subbottom3.

--set-source-camera=*sensorid*
Set the specified sensor to be a source for camera.

--set-source-camera1=*sensorid*
Set the specified sensor to be a source for camera1.

--set-source-camera2=*sensorid*
Set the specified sensor to be a source for camera2.

--set-source-camera3=*sensorid*
Set the specified sensor to be a source for camera3.

--set-source-position=*sensorid*
Set the specified sensor to be a source for position.

--set-source-position1=*sensorid*
Set the specified sensor to be a source for position1.

--set-source-position2=*sensorid*
Set the specified sensor to be a source for position2.

--set-source-position3=*sensorid*
Set the specified sensor to be a source for position3.

--set-source-depth=*sensorid*
Set the specified sensor to be a source for depth.

--set-source-depth1=*sensorid*
Set the specified sensor to be a source for depth1.

--set-source-depth2=*sensorid*
Set the specified sensor to be a source for depth2.

--set-source-depth3=*sensorid*
Set the specified sensor to be a source for depth3.

--set-source-heading=*sensorid*
Set the specified sensor to be a source for heading.

--set-source-heading1=*sensorid*
Set the specified sensor to be a source for heading1.

--set-source-heading2=*sensorid*
Set the specified sensor to be a source for heading2.

--set-source-heading3=*sensorid*
Set the specified sensor to be a source for heading3.

--set-source-rollpitch=*sensorid*
Set the specified sensor to be a source for rollpitch.

--set-source-rollpitch1=*sensorid*
Set the specified sensor to be a source for rollpitch1.

--set-source-rollpitch2=*sensorid*
Set the specified sensor to be a source for rollpitch2.

--set-source-rollpitch3=*sensorid*
Set the specified sensor to be a source for rollpitch3.

--set-source-heave=*sensorid*
Set the specified sensor to be a source for heave.

--set-source-heave1=*sensorid*
Set the specified sensor to be a source for heave1.

--set-source-heave2=*sensorid*
Set the specified sensor to be a source for heave2.

--set-source-heave3=*sensorid*
Set the specified sensor to be a source for heave3.

]

EXAMPLES

The Monterey Bay Aquarium Research Institute operates a Low Altitude Survey System (LASS) that is mounted on Remotely Operated Vehicles (ROVs) and used for 1-cm-scale surveys of the seafloor. The LASS combines the following sensors:

Kearfott SeaDevil inertial navigation system (INS)
Paroscientific Digiquartz pressure sensor
Teledyne Reson T50 400 kHz multibeam sonar
3D at Depth Wide Swath Subsea Lidar (WiSSL)
Stereo camera rig with two Allied Vision Prosilica GX1920 color cameras
VectorNav VN100 inertial measurement unit (IMU) and attitude heading reference system (AHRS)

MB-System processing of the various datasets collected by the LASS depend upon the platform model for the LASS, which is created by the following set of commands to **mbmakeplatform**:

```
mbmakeplatform --output=20240829ROVVentana.plf \
    --platform-type-rov \
    --platform-name="Ventana" \
    --platform-organization="Monterey Bay Aquarium Research Institute" \
    --add-sensor-ins \
```

```
    --sensor-model="SeaDeViL" \
    --sensor-manufacturer="Kearfott" \
    --sensor-serialnumber="2" \
    --sensor-capability-position \
    --sensor-capability-heading \
    --sensor-capability-rollpitch \
    --sensor-offsets=0.0/0.0/0.0/0.0/0.0/0.0 \
    --sensor-time-latency=0.0 \
    --sensor-source-position \
    --sensor-source-heading \
    --sensor-source-rollpitch \
    --sensor-source-heave \
--end-sensor \
--add-sensor-pressure \
    --sensor-model="Digiquartz" \
    --sensor-manufacturer="Paroscientific" \
    --sensor-serialnumber="Unknown" \
    --sensor-capability-depth \
    --sensor-offset-positions=-0.2164694/-0.076170/-0.069085 \
    --sensor-time-latency=0.0 \
    --sensor-source-depth \
--end-sensor \
--add-sensor-sonar-multibeam \
    --sensor-model="T50 400 kHz" \
    --sensor-manufacturer="Reson" \
    --sensor-serialnumber="Unknown" \
    --sensor-capability-topography-multibeam \
    --sensor-capability-backscatter-multibeam \
    --sensor-offsets=-0.459100/0.246774/-0.180388/0.07/0.07/0.16 \
    --sensor-offsets=-0.459100/0.442174/-0.211386/0.07/0.07/0.16 \
    --sensor-source-bathymetry \
    --sensor-source-bathymetry1 \
    --sensor-source-backscatter \
    --sensor-source-backscatter1 \
--end-sensor \
--add-sensor-lidar-swath \
    --sensor-model="WiSSL" \
    --sensor-manufacturer="3DatDepth" \
    --sensor-serialnumber="1" \
    --sensor-capability-topography-lidar \
    --sensor-capability-backscatter-lidar \
    --sensor-offsets=-0.4653/0.0160/-0.0952/0.0/0.0/0.0 \
    --sensor-source-bathymetry2 \
    --sensor-source-backscatter2 \
--end-sensor \
--add-sensor-camera-stereo \
    --sensor-model="Prosilica GX1920" \
    --sensor-manufacturer="Allied Vision" \
    --sensor-serialnumber="Unknown" \
    --sensor-capability-stereophotography \
    --sensor-offsets=-0.559430/-0.106901/-0.136561/0.0/0.0/0.0 \
    --sensor-offsets=-0.559430/-0.106901/-0.136561/0.0/0.0/0.0 \
    --sensor-source-bathymetry3 \
```

```
    --sensor-source-backscatter3 \
  --end-sensor \
  --add-sensor-ins \
    --sensor-model="VN-100" \
    --sensor-manufacturer="VectorNav" \
    --sensor-serialnumber="Unknown" \
    --sensor-capability-position \
    --sensor-capability-heading \
    --sensor-capability-rollpitch \
    --sensor-offsets=--0.255900/0.231999/-0.004164/0.0/0.0/0.0 \
    --sensor-time-latency=0.0 \
    --sensor-source-position \
    --sensor-source-heading \
    --sensor-source-rollpitch \
    --sensor-source-heave \
  --end-sensor
```

Executing the above command produces a platform file named 20240829ROVVentana.plf, which has the following contents:

```
## MB-System Platform Definition File
MB-SYSTEM_VERSION       5.8.2beta13
FILE_VERSION            1.00
ORIGIN                  Generated by user <seafloor> on cpu <Morgan.local> at <Thu Aug 29 14:33:58
2024>
##
PLATFORM_TYPE           3  ## ROV
PLATFORM_NAME           Ventana
PLATFORM_ORGANIZATION   Monterey Bay Aquarium Research Institute
DOCUMENTATION_URL
##
START_TIME_D            0.000000  ## 0000/00/00 00:00:00.000000
END_TIME_D              0.000000  ## 0000/00/00 00:00:00.000000
##
PLATFORM_NUM_SENSORS    6
##
## Defined data source sensors:
SOURCE_BATHYMETRY       2
SOURCE_BATHYMETRY1      2
SOURCE_BATHYMETRY2      3
SOURCE_BATHYMETRY3      4
SOURCE_BACKSCATTER      2
SOURCE_BACKSCATTER1     2
SOURCE_BACKSCATTER2     3
SOURCE_BACKSCATTER3     4
SOURCE_POSITION         5
SOURCE_DEPTH            1
SOURCE_HEADING          5
SOURCE_ROLLPITCH        5
SOURCE_HEAVE            5
##
## Undefined data sources:
 ## SOURCE_SUBBOTTOM
```

```
        ## SOURCE_SUBBOTTOM1
        ## SOURCE_SUBBOTTOM2
        ## SOURCE_SUBBOTTOM3
        ## SOURCE_CAMERA
        ## SOURCE_CAMERA1
        ## SOURCE_CAMERA2
        ## SOURCE_CAMERA3
        ## SOURCE_POSITION1
        ## SOURCE_POSITION2
        ## SOURCE_POSITION3
        ## SOURCE_DEPTH1
        ## SOURCE_DEPTH2
        ## SOURCE_DEPTH3
        ## SOURCE_HEADING1
        ## SOURCE_HEADING2
        ## SOURCE_HEADING3
        ## SOURCE_ROLLPITCH1
        ## SOURCE_ROLLPITCH2
        ## SOURCE_ROLLPITCH3
        ## SOURCE_HEAVE1
        ## SOURCE_HEAVE2
        ## SOURCE_HEAVE3
      ##
      ## Sensor list:
      ##
      SENSOR_TYPE             0  101  ## INS
      SENSOR_MODEL            0  SeaDeViL
      SENSOR_MANUFACTURER     0  Kearfott
      SENSOR_SERIALNUMBER     0  2
      SENSOR_CAPABILITY1      0      193  ## position rollpitch heading heading
      SENSOR_CAPABILITY2      0      0  ##
      SENSOR_NUM_OFFSETS      0  1
      OFFSET_POSITION         0      0    0.000000    0.000000    0.000000 ## Starboard, Forward, Up
(meters)
      OFFSET_ATTITUDE         0      0    0.000000    0.000000    0.000000 ## Heading, Roll, Pitch
(degrees)
      SENSOR_TIME_LATENCY_STATIC  0      0.000000  ## Seconds
      ##
      SENSOR_TYPE             1  111  ## Pressure
      SENSOR_MODEL            1  Digiquartz
      SENSOR_MANUFACTURER     1  Paroscientific
      SENSOR_SERIALNUMBER     1  Unknown
      SENSOR_CAPABILITY1      1      2  ## depth
      SENSOR_CAPABILITY2      1      0  ##
      SENSOR_NUM_OFFSETS      1  1
      OFFSET_POSITION         1      0   -0.216469   -0.076170   -0.069085 ## Starboard, Forward, Up
(meters)
      SENSOR_TIME_LATENCY_STATIC  1      0.000000  ## Seconds
      ##
      SENSOR_TYPE             2   30  ## Sonar multibeam
      SENSOR_MODEL            2  T50 400 kHz
      SENSOR_MANUFACTURER     2  Reson
      SENSOR_SERIALNUMBER     2  Unknown
```

```
        SENSOR_CAPABILITY1          2          0  ##
        SENSOR_CAPABILITY2          2      65544  ## topography_multibeam backscatter_multibeam
        SENSOR_NUM_OFFSETS          2   2
        OFFSET_POSITION             2      0   -0.459100    0.246774   -0.180388 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             2      0    0.070000    0.070000    0.160000 ## Heading, Roll, Pitch
(degrees)
        OFFSET_POSITION             2      1   -0.459100    0.442174   -0.211386 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             2      1    0.070000    0.070000    0.160000 ## Heading, Roll, Pitch
(degrees)
        ##
        SENSOR_TYPE             3   61  ## Lidar swath
        SENSOR_MODEL            3  WiSSL
        SENSOR_MANUFACTURER         3  3DatDepth
        SENSOR_SERIALNUMBER         3  1
        SENSOR_CAPABILITY1          3          0  ##
        SENSOR_CAPABILITY2          3     131136  ## topography_lidar backscatter_lidar
        SENSOR_NUM_OFFSETS          3   1
        OFFSET_POSITION             3      0   -0.465300    0.016000   -0.095200 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             3      0    0.000000    0.000000    0.000000 ## Heading, Roll, Pitch
(degrees)
        ##
        SENSOR_TYPE             4   51  ## Camera stereo
        SENSOR_MODEL            4  Prosilica GX1920
        SENSOR_MANUFACTURER         4  Allied Vision
        SENSOR_SERIALNUMBER         4  Unknown
        SENSOR_CAPABILITY1          4          0  ##
        SENSOR_CAPABILITY2          4   33554432  ## stereophotography
        SENSOR_NUM_OFFSETS          4   2
        OFFSET_POSITION             4      0   -0.559430   -0.106901   -0.136561 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             4      0    0.000000    0.000000    0.000000 ## Heading, Roll, Pitch
(degrees)
        OFFSET_POSITION             4      1   -0.559430   -0.106901   -0.136561 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             4      1    0.000000    0.000000    0.000000 ## Heading, Roll, Pitch
(degrees)
        ##
        SENSOR_TYPE             5  101  ## INS
        SENSOR_MODEL            5  VN-100
        SENSOR_MANUFACTURER         5  VectorNav
        SENSOR_SERIALNUMBER         5  Unknown
        SENSOR_CAPABILITY1          5        193  ## position rollpitch heading heading
        SENSOR_CAPABILITY2          5          0  ##
        SENSOR_NUM_OFFSETS          5   1
        OFFSET_POSITION             5      0    0.000000    0.000000    0.000000 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             5      0    0.000000    0.000000    0.000000 ## Heading, Roll, Pitch
(degrees)
        SENSOR_TIME_LATENCY_STATIC  5       0.000000  ## Seconds
        ##
```

The Monterey Bay Aquarium Research Institute operates two Dorado class autonomous underwater vehicles (AUVs) optimized for seafloor mapping in the deep ocean. These Mapping AUVs field the following sensors:

Kearfott SeaDevil inertial navigation system (INS)
Paroscientific Digiquartz pressure sensor
Teledyne Reson T50 400 kHz multibeam sonar
Edgetech 110 kHz chirp sidescan sonar
Edgetech 1-6 kHz chirp subbottom profiler

MB-System processing of the various datasets collected by the Mapping AUVs depend upon the platform models for these vehicles, which are created by the following set of commands to **mbmakeplatform**:

```
mbmakeplatform --output=20240510m1_MAUV2.plf \
        --platform-type-auv \
        --platform-name="MAUV2 MBARI Dorado Class Mapping AUV" \
        --platform-organization="Monterey Bay Aquarium Research Institute" \
        --add-sensor-ins \
           --sensor-model="SeaDeViL" \
           --sensor-manufacturer="Kearfott" \
           --sensor-serialnumber="2" \
           --sensor-capability-position \
           --sensor-capability-heading \
           --sensor-capability-rollpitch \
           --sensor-offsets=0.0/0.0/0.0/0.0/0.0/0.0 \
           --sensor-time-latency=0.0 \
           --sensor-source-position \
           --sensor-source-heading \
           --sensor-source-rollpitch \
           --sensor-source-heave \
        --end-sensor \
        --add-sensor-pressure \
           --sensor-model="Digiquartz" \
           --sensor-manufacturer="Paroscientific" \
           --sensor-serialnumber="Unknown" \
           --sensor-capability-depth \
           --sensor-offset-positions=0.0/-0.6/0.0 \
           --sensor-time-latency=0.0\
           --sensor-source-depth \
        --end-sensor \
        --add-sensor-sonar-multibeam \
           --sensor-model="7125 400 kHz" \
           --sensor-manufacturer="Reson" \
           --sensor-serialnumber="Unknown" \
           --sensor-capability-topography-multibeam \
           --sensor-capability-backscatter-multibeam \
           --sensor-offsets=0.0/0.353/-0.18/0.50/-0.16/+0.34 \
          --sensor-offsets=0.0/0.153/+0.00/0.50/-0.16/+0.34 \
           --sensor-source-bathymetry \
           --sensor-source-bathymetry1 \
           --sensor-source-backscatter \
           --sensor-source-backscatter1 \
        --end-sensor \
        --add-sensor-sonar-sidescan \
           --sensor-model="FSAU chirp 110 kHz" \
```

```
    --sensor-manufacturer="Edgetech" \
    --sensor-serialnumber="Unknown" \
    --sensor-capability-backscatter-sidescan \
    --sensor-offsets=0.0/1.353/-0.10/0.0/0.0/0.0 \
    --sensor-source-backscatter2 \
  --end-sensor \
  --add-sensor-sonar-subbottom \
    --sensor-model="FSAU chirp 1-6 kHz" \
    --sensor-manufacturer="Edgetech" \
    --sensor-serialnumber="Unknown" \
    --sensor-capability-backscatter-sidescan \
    --sensor-offsets=0.0/1.353/-0.10/0.0/0.0/0.0 \
    --sensor-source-backscatter2 \
  --end-sensor
```

Executing the above command produces a platform file named 20240510m1_MAUV2.plf, which has the following contents:

```
## MB-System Platform Definition File
MB-SYSTEM_VERSION       5.8.2beta02
FILE_VERSION            1.00
ORIGIN                  Generated by user <caress> on cpu <auvrouter.rc.mbari.org> at <Sun May 12 20:48:01 2024>
##
PLATFORM_TYPE           4  ## AUV
PLATFORM_NAME           MAUV2 MBARI Dorado Class Mapping AUV
PLATFORM_ORGANIZATION   Monterey Bay Aquarium Research Institute
DOCUMENTATION_URL
##
START_TIME_D            0.000000  ## 0000/00/00 00:00:00.000000
END_TIME_D              0.000000  ## 0000/00/00 00:00:00.000000
##
PLATFORM_NUM_SENSORS    5
##
## Defined data source sensors:
SOURCE_BATHYMETRY       2
SOURCE_BATHYMETRY1      2
SOURCE_BACKSCATTER      2
SOURCE_BACKSCATTER1     2
SOURCE_BACKSCATTER2     4
SOURCE_POSITION         0
SOURCE_DEPTH            1
SOURCE_HEADING          0
SOURCE_ROLLPITCH        0
SOURCE_HEAVE            0
##
## Undefined data sources:
 ## SOURCE_BATHYMETRY2
 ## SOURCE_BATHYMETRY3
 ## SOURCE_BACKSCATTER3
 ## SOURCE_SUBBOTTOM
 ## SOURCE_SUBBOTTOM1
```

```
        ## SOURCE_SUBBOTTOM2
        ## SOURCE_SUBBOTTOM3
        ## SOURCE_CAMERA
        ## SOURCE_CAMERA1
        ## SOURCE_CAMERA2
        ## SOURCE_CAMERA3
        ## SOURCE_POSITION1
        ## SOURCE_POSITION2
        ## SOURCE_POSITION3
        ## SOURCE_DEPTH1
        ## SOURCE_DEPTH2
        ## SOURCE_DEPTH3
        ## SOURCE_HEADING1
        ## SOURCE_HEADING2
        ## SOURCE_HEADING3
        ## SOURCE_ROLLPITCH1
        ## SOURCE_ROLLPITCH2
        ## SOURCE_ROLLPITCH3
        ## SOURCE_HEAVE1
        ## SOURCE_HEAVE2
        ## SOURCE_HEAVE3
      ##
      ## Sensor list:
      ##
      SENSOR_TYPE             0  101  ## INS
      SENSOR_MODEL            0  SeaDeViL
      SENSOR_MANUFACTURER     0  Kearfott
      SENSOR_SERIALNUMBER     0  2
      SENSOR_CAPABILITY1      0      193  ## position rollpitch heading heading
      SENSOR_CAPABILITY2      0      0  ##
      SENSOR_NUM_OFFSETS      0  1
      OFFSET_POSITION         0      0    0.000000    0.000000    0.000000 ## Starboard, Forward, Up
(meters)
      OFFSET_ATTITUDE         0      0    0.000000    0.000000    0.000000 ## Heading, Roll, Pitch
(degrees)
      SENSOR_TIME_LATENCY_STATIC  0      0.000000  ## Seconds
      ##
      SENSOR_TYPE             1  111  ## Pressure
      SENSOR_MODEL            1  Digiquartz
      SENSOR_MANUFACTURER     1  Paroscientific
      SENSOR_SERIALNUMBER     1  Unknown
      SENSOR_CAPABILITY1      1      2  ## depth
      SENSOR_CAPABILITY2      1      0  ##
      SENSOR_NUM_OFFSETS      1  1
      OFFSET_POSITION         1      0    0.000000   -0.600000    0.000000 ## Starboard, Forward, Up
(meters)
      SENSOR_TIME_LATENCY_STATIC  1      0.000000  ## Seconds
      ##
      SENSOR_TYPE             2   30  ## Sonar multibeam
      SENSOR_MODEL            2  7125 400 kHz
      SENSOR_MANUFACTURER     2  Reson
      SENSOR_SERIALNUMBER     2  Unknown
      SENSOR_CAPABILITY1      2      0  ##
```

```
        SENSOR_CAPABILITY2          2       65544  ## topography_multibeam backscatter_multibeam
        SENSOR_NUM_OFFSETS          2   2
        OFFSET_POSITION             2       0    0.000000    0.353000   -0.180000 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             2       0    0.500000   -0.160000    0.340000 ## Heading, Roll, Pitch
(degrees)
        OFFSET_POSITION             2       1    0.000000    0.153000    0.000000 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             2       1    0.500000   -0.160000    0.340000 ## Heading, Roll, Pitch
(degrees)
        ##
        SENSOR_TYPE             3   20  ## Sonar sidescan
        SENSOR_MODEL            3  FSAU chirp 110 kHz
        SENSOR_MANUFACTURER         3  Edgetech
        SENSOR_SERIALNUMBER         3  Unknown
        SENSOR_CAPABILITY1          3         0  ##
        SENSOR_CAPABILITY2          3      8192  ## backscatter_sidescan
        SENSOR_NUM_OFFSETS          3   1
        OFFSET_POSITION             3       0    0.000000    1.353000   -0.100000 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             3       0    0.000000    0.000000    0.000000 ## Heading, Roll, Pitch
(degrees)
        ##
        SENSOR_TYPE             4   40  ## Sonar subbottom
        SENSOR_MODEL            4  FSAU chirp 1-6 kHz
        SENSOR_MANUFACTURER         4  Edgetech
        SENSOR_SERIALNUMBER         4  Unknown
        SENSOR_CAPABILITY1          4         0  ##
        SENSOR_CAPABILITY2          4      8192  ## backscatter_sidescan
        SENSOR_NUM_OFFSETS          4   1
        OFFSET_POSITION             4       0    0.000000    1.353000   -0.100000 ## Starboard, Forward, Up
(meters)
        OFFSET_ATTITUDE             4       0    0.000000    0.000000    0.000000 ## Heading, Roll, Pitch
(degrees)
        ##
```

SEE ALSO

mbsystem(1), **mbpreprocess**(1), **mbprocess**(1)

BUGS

Platform files are complicated, yet inscrutable.